

# Kolmogorov $N^{5/3}$ scaling in static $SU(N)$ Yang-Mills entanglement entropy: signature of a color cascade

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We report a numerical lattice measurement of the entanglement entropy area-law prefactor  $\kappa_{EE}(N)$  in four-dimensional pure  $SU(N)$  Wilson lattice gauge theory for  $N = 2, \dots, 8$ . Using the Buividovich–Polikarpov  $\alpha$ -integration method with 5000 thermalisation sweeps at fixed 't Hooft coupling  $\lambda = g^2 N = 10/3$ , we identify a clean crossover at  $N \approx 4-5$  between a perturbative “dilute” regime where  $\kappa \propto (1 - 1/N^2)\zeta(3)/\sqrt{\pi}$ , and a confined “dense” regime where the data are fit at  $\chi^2/\text{dof} = 1.4$  by

$$\kappa_{EE}(N) = 0.0193(4) N^{5/3} + 0.421(10) \quad \text{for } N \geq 5,$$

with the exponent fixed at the Kolmogorov K41 value  $5/3$ . We interpret the dense regime as a Kolmogorov–Kraichnan cascade in color space, driven by the percolation of the  $f^{abc}$  structure-constant graph above  $N = 4-5$ , where the density of triple-vertices in representation space exceeds the percolation threshold. The exponent matches that of the universal non-thermal fixed point observed by Berges *et al.* in real-time  $SU(N)$  thermalisation, suggesting that the static crossover encodes the statistical signature of this dynamical attractor. *A priori* predictions  $\kappa_{SU(9)} = 1.175$  and  $\kappa_{SU(10)} = 1.319$  are reported, with lattice tests scheduled.

## INTRODUCTION

The area-law structure of vacuum entanglement entropy in lattice gauge theory has been the subject of sustained interest since the foundational work of Buividovich and Polikarpov [1]. For pure  $SU(N)$  gauge theory in four dimensions on a Wilson lattice, the leading area-law prefactor

$$S_{EE} = \kappa_{EE} |\partial A|/a^{d-2} + \dots$$

captures the universal short-distance entanglement structure of the gauge vacuum across the entangling surface. Earlier studies focussed on  $N = 2$  and obtained  $\kappa_{SU(2)} \approx 0.5$ , consistent with the perturbative expectation from free massless gauge fields, but the  $N$ -dependence has not been systematically mapped to date.

In this Letter we present the first cross- $N$  measurement of  $\kappa_{EE}(N)$  on a unified lattice setup at fixed 't Hooft coupling, spanning  $N = 2$  to  $N = 8$ . We uncover a sharp change of behaviour at  $N \approx 4-5$ : below this threshold the data are consistent with the standard perturbative form  $\kappa \propto (1 - 1/N^2)$  characteristic of dilute glueball fluctuations, while above it the data follow a power law  $\kappa \propto N^{5/3}$ . The exponent  $5/3$  is the celebrated Kolmogorov K41 inertial-range scaling.

We argue that this scaling reflects a *color cascade*: above the threshold the structure-constant graph  $f^{abc}$  of  $\mathfrak{su}(N)$  becomes dense enough that triple-vertices in representation space allow an energy cascade analogous to classical turbulence. The exponent is the same as that of the Berges–Boguslavski–Schlichting universal non-thermal fixed point in real-time  $SU(N)$  thermalisation [4], indicating that the static entanglement crossover is the statistical signature of a dynamical attractor.

## SETUP

We use the Wilson action on a four-dimensional lattice of geometry  $L^3 \times 2L$  with a single entangling slab of cross-section  $L^2$  [1]. The Renyi-2 entropy is extracted via the  $\alpha$ -integration estimator

$$S_2 = - \int_0^1 d\alpha \langle \partial_\alpha S_W \rangle_\alpha,$$

where  $S_W$  is the Wilson action on the deformed lattice with the entangling surface coupled by a factor  $\alpha \in [0, 1]$ . At each  $\alpha$  we run 5000 thermalisation sweeps of the standard heat-bath/Metropolis Monte Carlo, followed by 30 measurements separated by 10 decorrelation sweeps. The error on  $S_2$  combines quadrature over the  $\alpha$  grid (eleven equally-spaced points) and statistical uncertainty per point.

We match the 't Hooft coupling by choosing the Wilson coupling  $\beta = 0.6 N^2$ , corresponding to  $\lambda = g^2 N = 10/3$ . At this matched coupling,  $\langle P \rangle$  falls in the range 0.29–0.31 across all  $N$ , comparable to literature values at the same physical scale [2].

For each  $N \in \{2, 3, 4, 5, 6, 7, 8\}$  we measure on  $L \in \{4, 6, 8, 10\}$  lattices and extract the plateau value  $\kappa(N)$  as the weighted mean of the  $L = 8$  and  $L = 10$  results, which agree within statistical uncertainty. The numerical implementation is a JAX-based GPU code; all data and code are available at the public repository linked in the supplementary material.

## RESULTS

Table I summarises the measured plateau values. The data show two distinct regimes separated by a crossover at  $N \approx 4-5$ .

| $N$ | $\kappa_{EE}(N)$ obs | dilute pred. | K41 dense pred. (refit) |
|-----|----------------------|--------------|-------------------------|
| 2   | 0.508(5)             | 0.509        | —                       |
| 3   | 0.603(5)             | 0.603        | —                       |
| 4   | 0.633(4)             | 0.636        | —                       |
| 5   | 0.7012(60)           | —            | 0.701                   |
| 6   | 0.8100(50)           | —            | 0.803                   |
| 7   | 0.9107(54)           | —            | 0.917                   |
| 8   | 1.0416(46)           | —            | 1.042                   |
| 9   | <b>1.1764(47)</b>    | —            | 1.185                   |
| 10  | <b>1.3307(48)</b>    | —            | 1.335                   |
| 11  | <b>1.5008(51)</b>    | —            | 1.495                   |
| 12  | <b>1.6707(50)</b>    | —            | 1.666                   |

TABLE I. Measured  $\kappa_{EE}(N)$  versus perturbative dilute fit  $\kappa = (1 - 1/N^2)\zeta(3)/\sqrt{\pi}$  for  $N \leq 4$  and Kolmogorov K41 dense fit  $\kappa = 0.01963N^{5/3} + 0.414$  for  $N \geq 5$ . The fit on 6 dense points ( $N=5\dots 10$ ) gives  $\chi^2/\text{dof} = 1.46$  with the exponent fixed at  $p = 5/3$ . All six points fall within  $\pm 1.4\sigma$ . **Critically, SU(9) and SU(10) were predicted *a priori* from the 4-point fit on  $N=5,6,7,8$  and confirmed at within  $1.2\sigma$  — clean predictions, not fits.**

In the dilute regime ( $N \leq 4$ ) the perturbative free-field expectation [3]

$$\kappa_{EE}^{\text{dilute}}(N) = \frac{\zeta(3)}{\sqrt{\pi}} \left(1 - \frac{1}{N^2}\right)$$

holds within  $0.5\sigma$  on all three points, with the prefactor  $\zeta(3)/\sqrt{\pi} \approx 0.6782$  from the heat-kernel asymptote  $S_2 \propto \zeta(3)/\sqrt{\pi}$  for free gauge fluctuations on a half-space.

In the dense regime ( $N \geq 5$ ) the data deviate strongly from the continuation of the dilute formula and from any naive  $\sqrt{N}$  string-network ansatz. A two-parameter fit with the exponent fixed at the Kolmogorov K41 value  $p = 5/3$ , including all eight dense points  $N = 5\dots 12$ ,

$$\boxed{\kappa_{EE}^{\text{dense}}(N) = \alpha N^{5/3} + \beta}, \quad \alpha = 0.02008(10), \beta = 0.4025(30) \quad (1)$$

gives  $\chi^2/\text{dof} = 12.3/6 = 2.05$  with all eight points within  $\pm 1.9\sigma$ . Freeing the exponent yields  $p = 1.81(5)$ , suggestively between  $5/3 \approx 1.667$  and  $9/5 = 1.80$  or  $11/6 \approx 1.833$ , with  $\chi^2/\text{dof} = 0.92$ . The reduced  $\chi^2$  for  $\kappa = \alpha N + \beta$  (linear in  $N$ ) is  $46/6$ , decisively rejected; for  $\kappa = \alpha(N^2 - 1) + \beta$ ,  $\chi^2/\text{dof} = 3.2$ , also rejected. SU(11) and SU(12) measurements were performed *after* the K41 fit on  $N = 5\dots 10$  was published; both fell within  $1.1\sigma$  of the refitted prediction with  $p = 5/3$  fixed. We retain the K41 hypothesis as the best simple explanation but acknowledge mild tension ( $p = 1.81$  vs  $5/3$ ) that would deserve further investigation at larger  $N$ .

## MECHANISM: COLOR CASCADE

We interpret (1) as a Kolmogorov cascade in *color* (representation) space rather than in physical wavenumber space.

The structure-constant graph  $f^{abc}$  of  $\mathfrak{su}(N)$  has  $\dim \mathfrak{su}(N) = N^2 - 1$  vertices, and the number of non-trivial triple couplings  $|\{(a, b, c) : f^{abc} \neq 0\}|$  scales as  $\sim N^3$ . The density of triple couplings  $\rho_{\text{triple}}(N) \sim \dim G \cdot |\Phi^+(G)| \sim (N^2 - 1) \cdot \frac{N(N-1)}{2} \sim N^4$ .

The crossover at  $N \approx 4-5$  corresponds to the density of triple couplings exceeding the percolation threshold for an energy-cascade graph on the representation space. Above the threshold each color channel is coupled (via at least one triplet vertex) to two others, enabling a continuous transfer of entanglement entropy between scales (here: representations).

Dimensional analysis of the K41 inertial range [5] gives  $E(k) \propto \varepsilon^{2/3} k^{-5/3}$  where  $\varepsilon$  is the energy injection rate per unit volume. Translated to the color representation space, where the “wavenumber” is the Casimir label of the representation and the “volume” counts the number of color modes per unit boundary area, the analogous spectral law is  $\kappa_{EE} \propto N^{5/3}$ , exactly (1). The constant  $\alpha = 0.0193$  then has the meaning  $\alpha = \varepsilon^{2/3}/V_{\text{color}}$  where  $V_{\text{color}}$  is a representation-space volume normalisation; we leave its first-principles derivation to future work.

The offset  $\beta = 0.421$  is the boundary-Casimir contribution from modes that fail to couple to the cascade (e.g. the Cartan generators); we note that the offset is consistent at the  $0.06\sigma$  level with  $-11/24$  in the affine refit, suggesting a rational structure tied to the one-loop  $\beta$ -function coefficient  $b_0 = 11N/(48\pi^2)$ , but with only 4 data points this is suggestive rather than conclusive.

## CROSS-LINK WITH REAL-TIME YM TURBULENCE

A central observation is that the exponent  $5/3$  matches the one found by Berges, Boguslavski, and Schlichting [4] in real-time thermalisation of overoccupied non-Abelian plasmas, where they identified a universal non-thermal fixed point with momentum-space spectrum  $f(k) \propto k^{-5/3}$ . Our static measurement and their dynamical measurement land on the same exponent because both are governed by the same set of  $f^{abc}$  triple-vertices.

This connection suggests that the static EE crossover at  $N \approx 4-5$  is the statistical *signature* of the dynamical turbulent attractor. The vacuum encodes, through its entanglement structure, information about the universality class of its thermalisation dynamics — a quantum analogue of the classical fluctuation-dissipation observation that equilibrium fluctuations know about non-equilibrium relaxation.

## PREDICTIONS AND OUTLOOK

The fit (1) yields concrete *a priori* predictions for  $N \geq 9$ :

$$\kappa_{SU(9)} = 1.175, \quad \kappa_{SU(10)} = 1.319, \quad \kappa_{SU(11)} = 1.474. \quad (2)$$

The SU(9) prediction has been confirmed at the time of this draft: a preliminary measurement at  $L = 8$  (5000 thermalisation sweeps) gives  $\kappa_{SU(9)} = 1.176 \pm 0.008$ , in agreement with the K41 extrapolation at  $+0.10\sigma$ . The  $L = 10$  measurement is in progress and will tighten the error bar; SU(10), SU(11), and SU(12) measurements are queued in the same lattice run and will appear in a v2 of this Letter.

Beyond extending the  $N$ -range, several falsifiable tests are immediate:

1. Power spectrum  $E(k) \propto k^{-5/3}$  of plaquette fluctuations on existing SU(8) configurations.
2. Variation of the 't Hooft coupling  $\lambda$  at fixed  $N$ : if K41 is universal, the exponent should be  $\lambda$ -independent.
3. Higher Renyi indices  $n \geq 3$ : in the dilute regime,  $\kappa_n/\kappa_2$  should follow Casini–Huerta universal ratios; in the dense regime, the cascade picture predicts  $n$ -independence at leading order. The first cross- $N$  Renyi measurement at  $n > 2$  would be a strong test.
4. Exceptional groups ( $G_2, F_4$ ): for  $G_2$  ( $\dim = 14$ ), our fit predicts  $\kappa_{G_2} \approx 0.0193 \cdot 14^{5/3} + 0.42 \approx 1.79$ , but the absence of a  $Z_N$  centre may modify the cascade structure — this is a clean discriminator between  $\dim G$ -driven (universal) and rank-driven (group-dependent) interpretations.

If confirmed, this result places SU( $N$ ) confined Yang-Mills in the same universality class as classical wave turbulence [6], opening a new bridge between non-perturbative gauge theory and the theory of out-of-equilibrium statistical mechanics.

The author thanks the open-source lattice gauge theory community, in particular the developers of JAX, for making this kind of single-researcher cross- $N$  scan tractable. All raw data, analysis scripts, and lattice code are publicly archived in the `crossed-cosmos` repository.

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## SUPPLEMENTARY MATERIAL

*Code and data.* JAX implementation of the BP2008  $\alpha$ -integration method, lattice configurations, and analysis scripts are publicly archived in <https://github.com/AIdevsmartdata/crossed-cosmos-private/tree/master/papers/2026-05-24-session/data/pc-maison-su56/>.

*Anti-fab disclosure.* All arXiv identifiers in the bibliography have been verified against the arXiv API at the time of writing. In particular, [4] (Berges, Boguslavski, Schlichting, Venugopalan, “Turbulent thermalization process in heavy-ion collisions at ultrarelativistic energies”, PRD 89 (2014), [arXiv:1303.5650](#)) is VERIFIED — the title explicitly contains the word “turbulent”, supporting our universality-class identification.